Homework

Question 22: *(From Hofmann, Ch. 3)* Let $f \in \mathbb{B}^+(Q)$ for some AFA's state set Q.

(a) Suppose $N \subseteq M \subseteq Q$ with $N \models f$. Show that $M \models f$ as well. Hint: use induction on the structure of positive Boolean formulae.

Proof. Positive Boolean formulae are defined recursively. We will take the base case of this recursive definition as our base case for induction, and subsequently the recursive cases as our inductive steps.

Base case: $f = q$ for some individual state $q \in Q$. By the definition of positive Boolean formulae, $N \models q \iff q \in N$. Since it is given that $N \models q$, we have $q \in N$. Finally, since $N \subseteq M$, we can conclude that $q \in M$.

Inductive step: $f = g \vee h$. We have given that $N \models g \vee h$. By the definition of positive Boolean formulae, then, $N \models g$ or $N \models h$. In either case, we arrive at the desired outcome by a single application of the base case.

Inductive step: $f = g \wedge h$. It is given that $N \models g \wedge h$, and therefore $N \models g$ and $N \models h$. Applying the base case once to each of these yields $M \models g$ and $M \models h$. Therefore, $M \models g \wedge h$. □

(b) Argue whether or not it is possible that $\varnothing \models f$ holds.

Proof. W.l.o.g., assume that f is an individual state. Then $\varnothing \models f \iff f \in \varnothing$. Clearly this statement is a contradiction, as $f \notin \varnothing$ for any f. Therefore, $\varnothing \not\models f$ for any f. □

(c) Argue whether or not it is possible that $Q \not\models f$ holds.

Proof. We have given that $f \in \mathbb{B}^+(Q)$. In other words, f is a positive Boolean formula over states of Q. Suppose $Q \not\models f$. Then by the definition of satisfaction, when all the states $q \in Q$ are set to *true* and all the states $q \notin Q$ are set to *false*, f evaluates to *false* under the usual rules of Boolean formula evaluation. But since f contains only states of Q and *positive* Boolean operators (*i.e.*, no negation), the truth of all states $q \in Q$ is enough to guarantee the truth of f. We can conclude that $Q \models f$ for all $f \in \mathbb{B}^+(Q)$. □

Question 23: Let $M \subseteq Q$ for some AFA's state set Q. We call M a *minimal model* of $f \in \mathbb{B}^+(Q)$ if $M \models f$ and $N \not\models f$ for all $N \subsetneq M$.

(a) Construct a positive Boolean formula that has more than one minimal model.

M is a minimal model of f if all states in M must be *true* for f to be *true*; in other words, if we make any state of M *false*, then f becomes *false* as well.

$M_1 = \{q_1\}$ and $M_2 = \{q_2\}$ are minimal models for the formula $f = q_1 \vee q_2$. It is clear from the definition of satisfaction that both M_1 and M_2 individually model f. Moreover, they are minimal models of f because their only proper subset is $\varnothing$, and we have already shown above in 22(b) that $\varnothing \not\models f$ for any f, and therefore in particular for $f = q_1 \vee q_2$.